



Oral Surgery

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“Pharmacology of Local Anesthesia” Part 2

Lecture 7

2.Vasoconstrictor

- **Vasoconstrictors** are drugs that constrict blood vessels thereby control tissue perfusion. They are added to the local anesthetic solutions to oppose the vasodilatation actions of the local anesthetic agent.
- After injection of the local anesthetic, blood vessels dilate in the area, resulting in an increased blood flow to the site. This increase in perfusion leads to the following reactions:

1. Increased rate of absorption of the local anesthetic into the cardiovascular system, which in turn removes it from the injection site.
2. Increased plasma level of the local anesthetic, with an increased risk of local anesthetic toxicity.
3. Decreased duration of action and decreased depth of anesthesia because it diffuses away from the injection site more rapidly.
4. Increased bleeding at the site of injection due to increased perfusion.

The advantages of additions of vasoconstrictors to the local anesthetic solution are:

1. Vasoconstrictors decrease blood flow (perfusion) to the site of the injection.
2. The absorption rate of the local anesthetics into the cardiovascular system is reduced, resulting in lower anesthetic plasma levels, so decrease the risk of systemic toxicity (lower the local anesthetic plasma levels).
3. Increase the duration of action of local anesthetics (higher volumes of the local anesthetic agent remain in and around the nerve for longer periods).
4. Decrease bleeding at the site of injection, especially useful when increase bleeding is anticipated (e.g. during a surgical procedure).

Dilution of vasoconstrictors:

The dilution of the vasoconstrictor is commonly referred to as ratio (1 to 1000, and is written as 1:1000). This 1:1000 means that there is 1 gm (1000 mg) of the drug contained in 1000 ml of solution.

Types of the vasoconstrictors:

1- Epinephrine (adrenaline) :-

Sources: it is either synthetic or obtained from adrenal medulla of animals.

Mode of action:-

It acts directly on both alpha α & beta β receptors

Systemic action:

a- Cardiovascular system

It causes increased systolic & diastolic B.P.

It causes increased heart rate & strength of contraction.

It causes increased stroke volume & cardiac output.

It causes increased myocardial O₂ consumption.

- On blood vessel it causes vasoconstriction & so frequently used alone as a vasoconstrictor for hemostasis during surgical procedures.

B-Respiratory system:-

Epinephrine is a **potent dilator of the smooth muscle of the bronchiole**; it is the drug of choice for management of acute asthma.

C- Central nervous system (CNS)

In the usual therapeutic dosage epinephrine **is not a potent CNS stimulant**. CNS stimulation occurs when an excessive dosage is given.

Availability in dentistry

Epinephrine is the most potent & widely used vasoconstrictor in dentistry. It is available in 1:50 000, 1:80 000, 1:100 000, 1:200 000 dilution.

Maximum dosage:

a- for pain control

The least concentrated solution that produces effective pain control should be used.

For normal healthy patient:

The max. dose is **0.2 mg** per appointment & this means:

10 ml of a 1:50 000 dilution (5 cartridge).

20 ml of a 1:100 000 dilution (**11 cartridge**).

40 ml of a 1:200 000 dilution (22 cartridge).

For patient with clinically significant cardiovascular disease:

The max. dose **0.04 mg** per appointment, that means:

2 ml of a 1:50 000 (1 cartridge).

4 ml of a 1:100 000 (**2 cartridge**).

8 ml of a 1:200 000 (4 cartridge).

b- For hemostasis:-

Epinephrine containing LA solution is used via infiltration into the site of operation to prevent or minimize bleeding during surgical procedures.

For normal healthy patient the 1:50 000 dilution of epinephrine is more effective in hemostasis than less concentrated 1:100 000, or 1:200 000.

For patient with cardiovascular dis. The dilution of 1:100 000 is considered the best.

2- Norepinephrine (noradrenalin)

Sources: either synthetic or obtained from adrenal medulla of animals.

Mode of action:

The mode of action is **almost exclusively on alpha receptors**. It also stimulates beta (β) receptors in the heart. Norepinephrine is one fourth (25%) as potent as epinephrine.

Respiratory system

Norepinephrine **does not relax smooth muscle as do epinephrine** & it is not clinically effective in the management of bronchospasm.

CNS

In the usual therapeutic dosage norepinephrine is not potent CNS stimulant CNS stimulating action predominate when an excessive dosage is given.

Availability in dentistry:

Sometimes, norepinephrine is used with LA as vasoconstrictor in a 1: 30 000 dilution.

Levonordefrin (Neo-cobefrin):

It is synthetic substance.

It acts through direct alpha receptor stimulation with some B beta activity. It produces less cardiac & CNS stimulation than epinephrine does. It is mainly used with mepivacaine in a 1:20 000 dilution. The max. dose for all patients should be 1 mg per appointment that mean 20 ml of a 1:20 000 dilution (11 cartridge).

Felypressin (octapressin):

Sources: it is the analogue of the antidiuretic hormone vasopressin .it is a non-sympathomimetic drug &it is categorized as vasoconstrictor.

Mode of action:

It acts as direct stimulant of vascular smooth muscle. Its action appears to be more pronounced on the venous than the arteriolar microcirculation.

Clinical application:

It is used as vasoconstrictor to decrease the absorption as well as to increase the duration of the anesthesia, it is **not recommended for use when hemostasis is required** because of their predominant effect on the venous rather than the arterial circulation.

Availability in dentistry:

It is used in a dilution of 0.03 iu/ml with 3% prilocaine (it means international unit). Max. dose is 0.27 IU, that means: 9ml of 0.03 iu/ml.

Selection of a vasoconstrictor

The selection of an appropriate vasoconstrictor is based on the following factors:

1. Length of the surgical procedure: The duration of plain anesthesia only lasts for few minutes; the addition of adrenaline will prolong the duration.
2. The requirement of hemostasis: Some of the vasoconstrictors are effective in minimizing the blood loss during the surgical procedure as adrenaline, while Felypressin is of minimal value in achieving hemostasis.

3. The requirement of postoperative pain control: Adding vasoconstrictor will prolong the pain-free status.

4. Physical status of the patient and medication used: The benefit and risks of including a vasoconstrictor in a local anesthetic solution in patients who are medically compromised must be weighed against benefits and risks of using plain anesthesia.

For the following group of patients, the use of local anesthetic agents with vasoconstrictor is **contraindicated**:

A. Patients with a significant cardiovascular disease such as ischemic heart disease, hypertension or cerebral strokes.

B. Patients with thyrotoxicosis.

C. Patients receiving monoamine-oxidase inhibitors or tricyclic antidepressant.

Calculation of the dose of local anesthetics in dentistry (Using lidocaine as an example)

As any other drug, LA has a maximum safe dose. The dose is calculated in mg. per kg. of body weight (mg/kg body weight) .

The amount of LA agent (e.g. lidocaine) that is present in the cartridge is written on the cartridge by its percent concentration.

The number of milligram of the agent in the cartridge therefore can be calculated by.:

multiplying the percent conc., (ex 2% - $2\text{gm}/100\text{ml}=2000\text{ mg}/100\text{ml}=20\text{ mg/ml}$) .

Thus 1.8 ml. cartridge of 2% lidocaine contains 36 mg lidocaine. (1.8×20)

Example:-

The maximum safe dose of lidocaine is 4.5 mg/kg body weight per appointment. This means that maximum dose of 2% lidocaine (without epinephrine)that could be given to a 70 kg patient about 300 mg. per appointment ($4.5 \times 70 = 315$ mg).

To calculate the max number of cartridge:

2% means 2 gm. /100 ml = 2000 mg/100 ml=20 mg/ml

20 mg /ml * 1.8 = 36 mg lidocaine inside a single cartridge.

300 mg / 36 mg =8 cartridges.

We should remember that the max. dose of the vasoconstrictor is 0.2 mg per appointment for adult healthy patient and 0.04 mg per appointment for patient with clinically significant cardiovascular disease.

